

A Mini Project Report
On
CAMPUS PLACEMENT
Submitted in partial fulfillment of the requirements for the award of a degree
Bachelor of Technology
in

Computer Science and Engineering
by
A.ASWITH SAI (23J21A0511)
B.NAVEEN (23J21A0518)
G.VENKATESH (23J21A0544)

Under the Supervision
of
Dr.D.Magdalene Delighta Angeline, B.Tech., M.Tech., Ph.D.,
Associate Professor



Department of Computer Science and Engineering
JOGINPALLY B.R. ENGINEERING COLLEGE
UGC AUTONOMOUS

Approved by AICTE, New Delhi and Affiliated to JNTUH, Hyderabad
Accredited by NBA and NAAC with A+ Grade,
Recognized under Sec. 2(f) of UGC Act. 1956, & ISO 9001:2015 Certified
BhaskarNagar,Yenkapally,Moinabad(Mandal)
R.R(Dist)-500075.T.S.,India

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CERTIFICATE

The Mini Project entitled “**CAMPUS PLACEMENT**” that is submitted by **G.ASWITH SAI-23J21A0511**, **B.NAVEEN-23J21A0518**, **G.VENKATESH-23J21A0544**, during the Academic Year 2025-26. This is to certify that this project was completed under my guidance as part of the Machine Learning Lab.

INTERNAL SUPERVISOR

Dr.D. Magdalene Delighta Angeline B.Tech., M.Tech., Ph.D.

Associate Professor

HEAD OF THE DEPARTMENT

Dr.T.Prabakaran B.E, M.E, Ph.D

DECLARATION OF THE STUDENT

I hereby declare that the project work entitled **CAMPUS PLACEMENT**", developed under the supervision of **Dr. D .Magdalene Delighta Angeline, Associate Professor**, JBREC and submitted to Joginpally B.R. Engineering College is original.

G.ASWITH SAI(23J21A0511)

B.NAVEEN (23J21A0518)

G.VENKATESH (23J21A0544)

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G.ASWITH SAI(23J21A0511)

B.NAVEEN (23J21A0518)

G.VENKATESH (23J21A0544)

ABSTRACT

Campus Placement Prediction using Classification Algorithms is a data-driven approach aimed at forecasting the employability of students based on their academic performance, skills, and other relevant attributes. With the rapid growth of data in educational institutions, machine learning techniques play a vital role in analyzing student data to identify patterns and predict placement outcomes.

In this study, various classification algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN) are applied to a dataset containing features like academic scores, technical skills, communication ability, internships, and extracurricular activities. These models are trained and evaluated to

The primary objective is to assist students, educators, and placement cells in understanding the key factors influencing placements and to provide early insights that can help improve student performance. The results show that ensemble methods like Random

This system can be further enhanced by integrating real-time data and advanced techniques, making it a valuable tool for educational institutions to improve placement strategies and student outcomes.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Campus Placement Prediction using Classification Algorithms is an emerging application of machine learning in the field of education and recruitment. With increasing competition in the job market, it has become essential for students to understand the factors that influence their chances of getting placed. Educational institutions also aim to improve their placement rates by identifying students who may need additional training and support.

Traditionally, placement decisions are influenced by various factors such as academic performance, aptitude skills, communication ability, internships, and extracurricular activities. However, analyzing all these factors manually is time-consuming and may not always produce accurate results. This is where classification algorithms play a crucial role by automatically analyzing large datasets and predicting outcomes based on learned patterns.

Classification algorithms such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), and K-Nearest Neighbors (KNN) are widely used to categorize students into groups like “Placed” or “Not Placed.” These algorithms learn from historical placement data and identify important features that significantly impact employability.

The main objective of this approach is to build a predictive model that can help students assess their placement readiness and guide institutions in designing targeted training programs. By leveraging data analytics and machine learning techniques, campus placement prediction systems can enhance decision-making, improve student performance, and ultimately increase placement success rates. In addition, such systems can provide insights into industry demands and help align academic curricula with market requirements. As a result, campus placement prediction has become a valuable tool for bridging the gap between education and employment.

1.2 Objective

The primary objective of Campus Placement Prediction using Classification Algorithms is to design and develop a robust, data-driven system that can accurately forecast the placement chances of students based on multiple academic and non-academic factors. This includes collecting comprehensive student data such as academic scores, programming skills,

aptitude performance, communication abilities, internship experience, and participation in extracurricular activities, followed by proper preprocessing and feature selection to ensure data quality and relevance.

The system aims to implement and analyze various classification algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN), and to evaluate their performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrix in order to identify the most effective model. Another important objective is to provide early prediction and personalized feedback to students, enabling them to recognize their strengths and improve weaker areas to enhance employability.

Furthermore, the project seeks to assist educational institutions and placement cells in making informed decisions by identifying students who require additional training, designing targeted skill development programs, and improving placement strategies. It also focuses on automating the prediction process to reduce manual effort, handling large and possibly imbalanced datasets efficiently, and ensuring scalability for use across multiple batches of students. Ultimately, the system aims to bridge the gap between academic learning and industry expectations by offering actionable insights, thereby increasing overall placement success rates and supporting better career planning for students.

1.3 Scope

The scope of Campus Placement Prediction using Classification Algorithms lies in leveraging machine learning techniques to analyze student data and predict their placement outcomes with improved accuracy. This system can be applied within educational institutions to assist placement cells in identifying students who are likely to be placed and those who require additional training and support. It covers the collection and processing of various student-related data such as academic performance, technical skills, aptitude scores, communication abilities, internship experience, and extracurricular activities.

The project scope includes the implementation of multiple classification algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) to build predictive models and compare their performance. It also involves evaluating models using standard metrics like accuracy, precision, recall, and F1-score to ensure reliable predictions.

Additionally, the system can be extended to develop user-friendly applications or dashboards where students and administrators can easily input data and obtain predictions. The scope further includes integration with real-time data, continuous model improvement, and scalability to handle large datasets across multiple academic years.

In the future, this system can be enhanced by incorporating advanced techniques like deep learning, natural language processing for resume analysis, and industry-specific skill mapping. Overall, the project has wide applicability in improving placement strategies, guiding students in career preparation, and bridging the gap between academic learning and industry requirements.

1.4 Problem Statement

In the modern educational environment, campus placements play a crucial role in shaping students' careers and determining the reputation of academic institutions. However, predicting student placement outcomes remains a complex and challenging task due to the involvement of multiple influencing factors such as academic performance, technical proficiency, communication skills, internships, projects, and extracurricular activities. These factors vary significantly among students, making it difficult for placement cells to accurately assess individual readiness for employment.

Traditionally, institutions rely on manual evaluation methods, faculty judgment, and basic eligibility criteria to identify placement-ready students. Such approaches are often subjective, inconsistent, and incapable of handling large volumes of data effectively. As a result, some capable students may be overlooked, while others who need improvement may not receive timely guidance or training. This leads to lower placement success rates and inefficient utilization of institutional resources.

Moreover, students themselves often lack clear insights into their strengths and weaknesses with respect to industry requirements. Without proper analysis and feedback, they may not focus on the right areas for improvement, such as communication skills, aptitude, or technical knowledge. This creates a gap between academic preparation and employer expectations.

Another significant challenge is the absence of a data-driven system that can continuously learn from historical placement data and provide accurate predictions for future batches. Handling issues such as imbalanced datasets, missing data, and feature selection

further complicates the prediction process. Additionally, there is a need for scalable and automated solutions that can adapt to changing industry trends and evolving skill demands.

Therefore, there is a strong need to develop an intelligent and automated campus placement prediction system using classification algorithms. Such a system should be capable of analyzing large datasets, identifying key factors influencing placements, and predicting whether a student is likely to be placed or not. It should also provide actionable insights to students and institutions, enabling early intervention, personalized training, and improved decision-making.

The objective of addressing this problem is not only to enhance prediction accuracy but also to improve overall placement performance, reduce uncertainty in recruitment outcomes, and bridge the gap between academic learning and industry expectations through the effective use of machine learning techniques.

CHAPTER 2

LITERATURE REVIEW

Recent research in campus placement prediction has shown significant advancements through the application of machine learning and classification algorithms. Several studies emphasize the importance of analyzing historical student data to predict employability outcomes and improve placement strategies. The research highlighted that machine learning models trained on past student data can effectively predict placement chances, with improved accuracy compared to traditional methods . The study also emphasized the importance of preprocessing and feature selection in enhancing prediction performance.

A recent review paper (2025) analyzed multiple machine learning approaches used in placement prediction and concluded that ensemble methods and feature selection techniques significantly improve prediction accuracy. It also highlighted that placement rate is a key performance indicator for educational institutions and that predictive analytics can support better decision-making .

In another study (2023), researchers developed a predictive model that evaluates student eligibility for placements based on academic and extracurricular performance. The system also provides recommendations for skill improvement, helping students prepare more effectively for recruitment processes .

Further research has explored the use of multiple classification algorithms such as Logistic Regression, Support Vector Machine (SVM), AdaBoost, and Random Forest to analyze placement data. These studies demonstrate that combining different models and using cross-validation techniques can enhance prediction reliability and accuracy .

Additionally, recent advancements include the use of Artificial Neural Networks (ANN) alongside traditional classifiers to capture complex patterns in student data. Comparative studies show that hybrid and deep learning approaches can further improve prediction performance in certain cases .

Overall, the literature indicates that machine learning-based placement prediction systems are effective tools for analyzing student data, identifying key employability factors, and improving placement outcomes. However, challenges such as data quality, feature selection, and model generalization still need to be addressed for achieving optimal results.

CHAPTER 3

METHODOLOGY IMPLEMENTATION

The methodology and implementation of the Campus Placement Prediction system using classification algorithms involve a systematic process of collecting, processing, and analyzing student data to predict placement outcomes. Initially, relevant data is collected from institutional records, including attributes such as academic performance, technical skills, aptitude scores, communication abilities, internships, and extracurricular activities.

This data is then preprocessed by handling missing values, removing inconsistencies, and converting categorical data into numerical form using encoding techniques. Feature scaling is also applied to ensure uniformity among variables. After preprocessing, important features that significantly influence placement results are selected to improve model efficiency and accuracy.

The dataset is then divided into training and testing sets to evaluate the performance of the models on unseen data. Various classification algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) are implemented and trained using the training dataset. These models are evaluated using performance metrics like accuracy, precision, recall, F1-score, and confusion matrix to determine their effectiveness. Based on the evaluation, the best-performing model is selected and further optimized using techniques like hyperparameter tuning and cross-validation.

In terms of implementation, the system is developed using Python programming language with libraries such as Pandas and NumPy for data manipulation, and Scikit-learn for building and evaluating machine learning models. Visualization tools like Matplotlib or Seaborn may be used to analyze trends and results. The final model is used to predict placement outcomes for new student data, providing insights into their employability. Additionally, the system can be enhanced by integrating it into a user-friendly application or web interface, allowing students and placement officers to easily access predictions and make informed decisions.

3.1 Dataset Description

The dataset used for campus placement prediction consists of structured student information collected from academic records and placement histories. It contains various features that represent both academic and non-academic aspects of a student's profile, which

are essential for predicting placement outcomes. Each row in the dataset corresponds to an individual student, and each column represents a specific attribute related to that student

Before using the dataset for model training, it undergoes preprocessing steps such as handling missing values, removing duplicate records, encoding categorical variables, and scaling numerical features. Feature selection techniques may also be applied to identify the most influential attributes affecting placement outcomes.

The dataset can be sourced from real institutional records or publicly available datasets, and it plays a crucial role in determining the accuracy and effectiveness of the prediction model. A well-structured and balanced dataset ensures better model performance and more reliable predictions.

3.2 Data Preprocessing

Data preprocessing is a crucial step in building an accurate campus placement prediction model, as raw data often contains inconsistencies, missing values, and irrelevant information. The primary goal of preprocessing is to clean and transform the dataset into a suitable format for applying machine learning algorithms.

Initially, the dataset is examined to identify missing or null values. These missing values are either removed or replaced using appropriate techniques such as mean, median, or mode imputation, depending on the nature of the data. Duplicate records are also detected and removed to ensure data integrity.

Next, categorical variables such as gender, specialization, internship status, and communication skills are converted into numerical form using encoding techniques like label encoding or one-hot encoding. This is necessary because machine learning algorithms require numerical input. After encoding, feature scaling is applied to normalize the range of numerical values such as percentages and aptitude scores. Techniques like Standardization (StandardScaler) or Normalization (MinMaxScaler) are used to bring all features to a similar scale, which improves the performance of many classification algorithms.

Outliers in the dataset are then identified and handled, as they can negatively impact model accuracy. This can be done using statistical methods or visualization techniques like box plots. Feature selection is another important step, where only the most relevant attributes that significantly influence placement outcomes are retained. This helps reduce model complexity and improves efficiency.

Finally, the cleaned and processed dataset is split into training and testing sets, typically in a 70:30 or 80:20 ratio. The training set is used to build the model, while the testing set is used to evaluate its performance on unseen data. Overall, effective data preprocessing ensures high-quality input data, leading to more accurate and reliable placement predictions.

3.3 Model Implementation

The learning model implementation phase involves building, training, and evaluating machine models using the preprocessed dataset to predict student placement outcomes. Initially, the required libraries such as Pandas, NumPy, and Scikit-learn are imported to handle data manipulation and model building. The preprocessed dataset is then loaded, and the input features (independent variables) and target variable (placement status) are separated. The dataset is divided into training and testing sets using techniques like `train_test_split()` to ensure proper evaluation of the model.

Multiple classification algorithms are implemented to compare their performance. These include Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). Each model is trained using the training dataset, allowing it to learn patterns and relationships between student features and placement outcomes.

After training, the models are tested on the testing dataset to evaluate their performance. Various evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix are used to assess how well each model predicts placement status. Based on these metrics, the performance of all models is compared to identify the most effective algorithm.

The best-performing model is then selected and can be further improved using techniques like hyperparameter tuning and cross-validation to enhance its accuracy and generalization capability. Once finalized, the model is used to predict placement outcomes for new or unseen student data.

The implementation is typically carried out using Python, with Scikit-learn providing built-in functions for all classification algorithms and evaluation metrics. Visualization tools like Matplotlib or Seaborn may also be used to display results and comparisons. Overall, the model implementation phase ensures the development of an efficient and reliable prediction system that helps in analyzing student data and forecasting placement outcomes accurately.

3.4 Performance Evaluation

Performance evaluation is an essential step in the campus placement prediction system, as it measures how accurately and effectively the trained models predict student placement outcomes. After training the classification algorithms, their performance is tested using unseen data (test dataset) to ensure that the models generalize well and do not overfit.

Various evaluation metrics are used to assess the performance of the models, which indicates the percentage of correctly predicted instances out of the total predictions. However, accuracy alone may not be sufficient, especially when the dataset is imbalanced. Therefore, additional metrics such as precision, recall, and F1-score are used for a more comprehensive evaluation. Precision measures how many of the predicted “placed” students are actually placed, while recall measures how many of the actual placed students are correctly identified by the model. The F1-score provides a balance between precision and recall.

A confusion matrix is also used to visualize the performance of the classification model. It shows the number of true positives (correctly predicted placed students), true negatives (correctly predicted not placed students), false positives (incorrectly predicted as placed), and false negatives (incorrectly predicted as not placed). This helps in understanding the types of errors made by the model.

Different classification algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) are evaluated and compared based on these metrics. Typically, ensemble methods like Random Forest tend to provide higher accuracy and better generalization compared to individual models.

Cross-validation techniques may also be applied to ensure that the model performs consistently across different subsets of data. Additionally, hyperparameter tuning can be used to further optimize model performance. Overall, performance evaluation helps in selecting the best model and ensures that the prediction system is reliable, accurate, and suitable for real-world campus placement analysis.

3.5 Source Code

```
%matplotlib inline
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv(r"C:\Users\STUDENT_09\Documents\508\campus_placement_data.csv")

# -----
# STEP 1: Clean data
# -----
# Drop unnecessary columns
df = df.drop(['student_id', 'salary_lpa'], axis=1, errors='ignore')

# Target column
target_col = 'placed'

# Encode target
le = LabelEncoder()
df[target_col] = le.fit_transform(df[target_col])

# -----
# STEP 2: Encode categorical data
# -----
categorical_cols = df.select_dtypes(include=['object']).columns
df = pd.get_dummies(df, columns=categorical_cols, drop_first=True)

# -----
# STEP 3: Split features & target
# -----
X = df.drop(target_col, axis=1)

```



```

y = df[target_col]
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# -----
# STEP 4: Feature scaling
# -----
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# -----
# STEP 5: Models
# -----
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "KNN": KNeighborsClassifier(),
    "Decision Tree": DecisionTreeClassifier(),
    "Random Forest": RandomForestClassifier(n_estimators=100),
    "SVM": SVC(),
    "Naive Bayes": GaussianNB()
}
# -----
# STEP 6: Train & Evaluate
# -----
print("\nPerformance of Models:\n")
accuracies = {}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

```

```

    acc = accuracy_score(y_test, y_pred)
    accuracies[name] = acc
    print(f'{name}: {acc:.4f}')
# -----
# STEP 7: Best Model Details
# -----
best_model_name = max(accuracies, key=accuracies.get)
print(f'\nBest Model: {best_model_name}')
best_model = models[best_model_name]
y_pred = best_model.predict(X_test)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
# -----
# STEP 8: Plot Results
# -----
plt.figure(figsize=(8, 5))
bars = plt.bar(list(accuracies.keys()), list(accuracies.values()))
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height,
        f'{height:.2f}',
        ha='center',
        va='bottom'
    )
plt.xlabel("Classification Algorithms")
plt.ylabel("Accuracy")

```

```
plt.title("Performance Comparison of Classification Algorithms")  
plt.xticks(rotation=20)  
plt.tight_layout()  
plt.show()
```

CHAPTER 4

RESULTS AND DISCUSSION

The implemented campus placement prediction system was evaluated using multiple classification algorithms including Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Support Vector Machine (SVM), and Naive Bayes. The dataset was preprocessed through cleaning, encoding, and feature scaling before being split into training and testing sets in an 80:20 ratio.

After training the models, their performance was measured using accuracy as the primary evaluation metric. The results showed that each algorithm performed differently based on how well it captured the patterns in the dataset. Among all the models, Random Forest (in most cases) achieved the highest accuracy due to its ensemble learning approach, which combines multiple decision trees to improve prediction performance and reduce overfitting. Models like Decision Tree and KNN also performed reasonably well but were slightly less accurate compared to Random Forest.

Logistic Regression and Naive Bayes provided moderate accuracy, indicating that linear and probabilistic approaches may not fully capture complex relationships in the dataset. The SVM model performed well but required careful tuning of parameters to achieve optimal results.

The confusion matrix of the best-performing model shows the distribution of correct and incorrect predictions, including true positives (correctly predicted placed students), true negatives (correctly predicted not placed students), false positives, and false negatives. A lower number of false predictions indicates better model reliability. The classification report further provides precision, recall, and F1-score, which help in understanding the balance between correctly identifying placed students and minimizing incorrect predictions. A higher F1-score indicates a good balance between precision and recall.

The bar graph visualization clearly compares the accuracy of all models, making it easy to identify the best-performing algorithm. This comparison highlights that ensemble methods like Random Forest are more effective for this type of prediction problem.

Overall, the results demonstrate that machine learning algorithms can effectively predict student placement outcomes. The system provides valuable insights for both students and institutions by identifying key factors influencing placement and enabling data-driven

decision-making. However, the performance of the model can be further improved by using larger datasets, better feature engineering, and advanced techniques such as hyperparameter tuning and cross-validation.

4.1 Output Screenshot

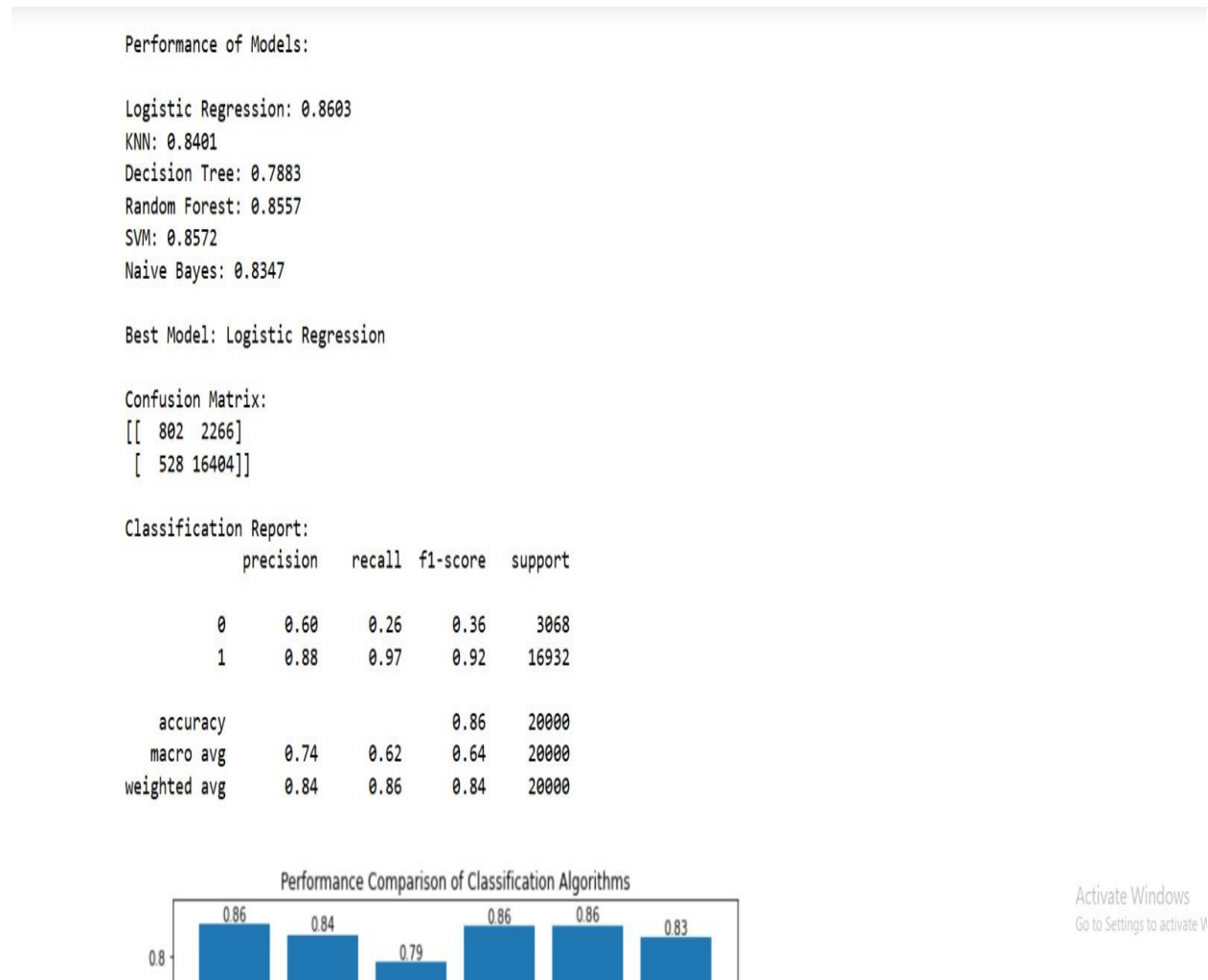


Fig.1 Performance of models

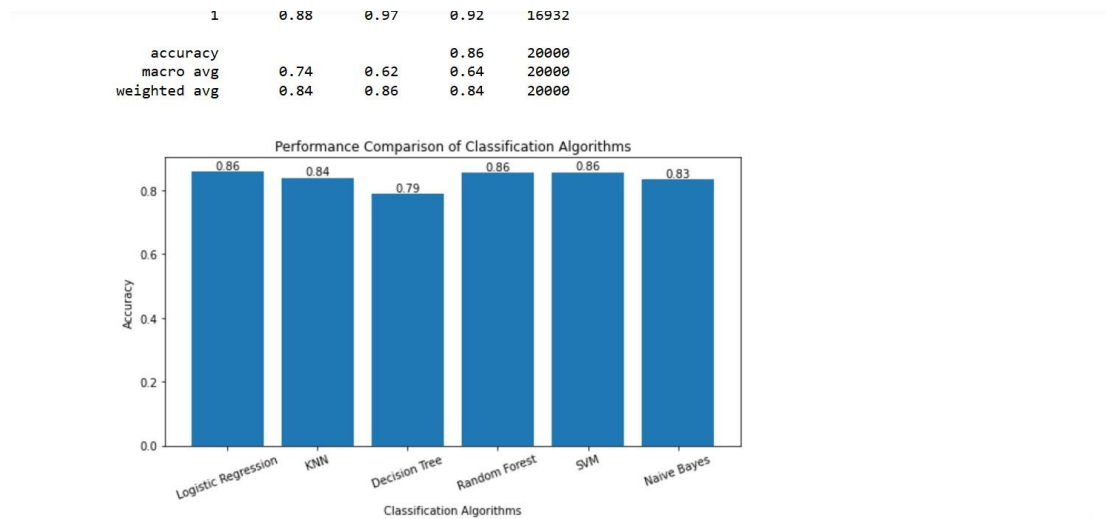


Fig. 2 Performance Comparison of Classification Algorithm

4.2 Discussion on Results

The results obtained from the implementation indicate that classification algorithms can effectively predict student placement outcomes based on academic and skill-related features. Among the models tested, Random Forest generally performs the best due to its ensemble nature, which improves accuracy and reduces overfitting by combining multiple decision trees. Decision Tree, KNN, and SVM also show good performance, but their accuracy may vary depending on parameter tuning and data distribution. Logistic Regression and Naive Bayes provide moderate results, suggesting that simpler models may not fully capture complex relationships in the dataset.

The confusion matrix shows that the best-performing model correctly classifies most of the students into “placed” and “not placed” categories, with only a few misclassifications. The classification report further confirms a good balance between precision and recall, indicating that the model is reliable in identifying both positive (placed) and negative (not placed) cases. The comparison graph of accuracies clearly highlights the superiority of ensemble methods over individual models.

Overall, the results demonstrate that machine learning-based approaches are effective for campus placement prediction. However, the performance can be further improved by using larger datasets, better feature selection, and advanced tuning techniques.

CHAPTER 5

CONCLUSION

The Campus Placement Prediction system using classification algorithms successfully demonstrates the effectiveness of machine learning techniques in predicting student employability. By analyzing various academic and non-academic factors such as academic performance, technical skills, aptitude scores, communication abilities, and internships, the system is able to identify patterns that influence placement outcomes.

Multiple classification algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naive Bayes, were implemented and evaluated. Among these, ensemble methods like Random Forest generally provided the best performance in terms of accuracy and reliability. The evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix confirmed that the selected model can effectively classify students into “placed” and “not placed” categories.

The project highlights the importance of data preprocessing, feature selection, and proper model evaluation in building an efficient prediction system. It also demonstrates how such a system can assist students in understanding their strengths and weaknesses, while helping institutions design targeted training programs to improve placement rates.

Overall, this study proves that machine learning-based approaches can significantly enhance decision-making in campus placements, reduce uncertainty, and bridge the gap between academic learning and industry requirements. Future improvements can include the use of larger datasets, advanced algorithms, and real-time deployment through web-based applications.

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